

Hyperscale AI Data Centres - Alberta Wilderness Association

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August 4, 2026 Joint Letter: Environment, climate, and community orgs call for a moratorium on AI Hyperscale Data Centre development On August 4, 2026, AWA, the Calgary Climate Hub, Climate Justice Edmonton, the Calgary-Philippines Solidarity... Hyperscale data centres are rapidly being proposed and developed throughout Alberta without sufficient public consultation and transparency. Hyperscale AI data centres have been found to reduce air and water quality, increase heat exposure and carbon emissions, cause noise pollution, and result in the loss of public lands, natural spaces, and agricultural areas. The public deserves to know the full scope and implications of these impacts. In December of 2024, the Government of Alberta announced its AI Data Centre Strategy, which sought to make Alberta “the North American destination of choice for AI data centre investment.” Since then, the government has been advertising Alberta’s deregulated electricity system, natural gas resources, fast-tracked regulatory approvals, and low corporate taxes to attract investment. The Alberta Electric Systems Operator (AESO) reports that requests from data centres to connect to the grid have “surged” since 2024 and continue to arrive. To put this in perspective, at the start of 2024, data centres had a total requested load of 200 MW on the grid. As of July 2026, over 20,000 MW of data centre load is on AESO’s project list. “If all of the proposed or under construction projects are completed, then Alberta would be home to 90 per cent of the country’s data centres.” – CBC News, June 4, 2026 Alberta is the epicentre of hyperscale data centre development in Canada, which is concerning considering that public input and consultation has been limited, with the governments of Alberta and Canada already allowing multiple data centres (Mihta Askiy, Synapse, and Wonder Valley) to bypass the Environmental Impact Assessments (EIA) process. The purpose of an EIA, in the provincial government’s own words, is to allow “companies and government decision makers to examine the effects that the proposed project may have on the environment and determine if the project is of public interest.” By bypassing this due process, government officials have decided on everyone’s behalf that AI data centres are in the public interest without any substantiating evidence. But research on existing data centres has found that their operations can have significant negative impacts. The computing power required by AI means massive energy demands. For example, operating Synapse Data Centre Inc., proposed in the town of Olds, would require 1,400 megawatts (MW) of power each day. It’s equivalent to the entirety of Edmonton’s daily use, and just one of dozens of similarly energy-intensive data centres hoping to operate in the province. The Alberta Electric Systems Operator (AESO) reports that requests from data centres to connect to the grid have “surged” since 2024 and “continue to arrive, with over 20,000 MW of data centre load now on the AESO project list.” To put this immense energy demand in perspective, the entire province

of Alberta is generating and consuming around 10,000 MW as this is being written. Based on AESO's last update in September of 2025, cumulative demand for power from data centres has increased 9900% percent since the beginning of 2024, with load well outpacing generation. As a pre-emptive measure, AESO has capped supply for data centres at 1200 MW and currently requires new operators to provide their own power rather than draw from the shared grid, stating that most data centres are "not contributing sufficient or timely generation, creating grid reliability issues". However, some projects are still pushing ahead and will be allowed to connect to the grid before their accompanying power plants are finished construction. Announced in early July of 2026, Meta's 970 MW data centre in Sturgeon County plans to start operating one to two years before construction of the Greenlight Electricity Centre, its neighbouring natural gas-fired power plant, is complete. Most proposed projects indicate similar intentions to construct large natural gas-based generators, which will increase air pollutants and greenhouse gas emissions, as well as compete with the public and other industries for the supply of natural gas within the province. The chemical compounds released when burning natural gas (largely comprised of methane) decrease air quality and are associated with increased short and long-term health burdens on the public, including respiratory illnesses, cardiovascular diseases, neurological disorders, poor reproductive outcomes, and premature mortality. Planned in the Municipal District of Greenview and purported to be the 'largest data centre on Earth,' Wonder Valley would need a 7,500 MW power plant to operate and require "about 10% of all gas supply in Alberta once fully operational." It would emit over 25 million tonnes of greenhouse gases alone, reversing all the progress the province has made on emissions reductions when coal power was phased out of the provincial grid. Access to Information requests have since revealed internal briefing notes from the federal government that promoted data centres as a way to boost production and create a new market for natural gas in Alberta. This backsliding has real costs, as intensifying carbon emissions contributes to climate change. Already, we are seeing the cascading effects that decades of climate change denial, dithering, and inaction have wrought. Alberta is located in one of the world's four "mega-drying regions," where terrestrial water storage is being lost at increasingly faster rates due to climate change. The landscape is losing its natural ability to retain water, which has profound implications for our livelihoods. A drier Alberta means increased water insecurity — less water reliably available for the economy and to meet all our society's essential needs (think health, hygiene, sanitation, safety, and emergency response). It means increased costs for food, as the loss of crop and livestock production squeezes supply and creates shortages, as well as higher energy costs, with extreme temperatures becoming more common. In this rapidly changing climate, droughts, wildfires, and floods all become more frequent, severe, and unpredictable, and with that comes increased expenses for responding to, managing, mitigating, and preventing the next natural disaster. "Water risk near data centre locations also varies significantly across provinces, with locations in Alberta and Saskatchewan displaying especially high rates of water risk." – Carlo & Rolheiser, 2026. AI data centres are known to consume considerable volumes of water for both cooling purposes and when generating electricity, which can contribute to local water scarcity, particularly in drought-prone regions. In addition to quantity, water quality can also be impacted, with research finding that the evaporative cooling processes often employed at data centres produce wastewater with high concentrations of contaminants. On the same day the Alberta government and Meta announced the development of a 970 MW data centre in Sturgeon County, it was reported Meta "flushed bacteria-contaminated water into public sewers during construction" of another data centre in Cheyenne, Wyoming. Many of the data centre developers in Alberta assert they will require limited water withdrawals or use 'closed loop systems,' but exact volumes, technologies, and management procedures remain vague. There also is seemingly little consideration of the increased water use and consumption from powering the data centres. Increasing natural gas supply requires more hydraulic fracturing, aka 'fracking', which injects water deep inside the earth, effectively taking it out of the provincial water balance for foreseeable generations. Fracking can also induce seismic activity and contaminate groundwater supplies. Assuming new combined cycle natural gas-fired turbines are constructed (as has been indicated by Meta and

Synapse's projects) and depending on the input fuel and cooling technology used, powering hyperscale data centres could consume anywhere between 833,000 (conventional gas, dry cooling) to 5.6 million (shale gas, cooling pond) litres of water annually per MW over their life cycle. This means that each year, a 1,000-MW plant would consume 833 million – 5.6 billion litres, the same amount of water as the annual use of 12,000 – 79,000 Albertans. If all 20,000+ MW requested by currently proposed data centres came to fruition, their power plants alone would consume as much water as 234,000 – 1.6 million Albertans each year. Water use intensity further increases if the power plants use carbon capture and storage (CCS) technologies to mitigate the climate-warming emissions their operations release; research finds that operational water consumption increases by 29 – 81% at combined cycle power plants with CCS, depending on cooling type. Adding all this new water demand and pressure is worrisome, considering that many watersheds, particularly in dry Southern Alberta, are already overallocated and intensely used; to continue to reliably provide to all those reliant on them, rivers like the Oldman and the Bow need more water flowing through them, not less. Other environmental issues associated with AI data centres include noise pollution, increased heat exposure, and loss or alteration of natural areas and agricultural lands and their associated benefits. At a time when we are already dealing with energy uncertainty, water shortages, climate change, ecosystem loss, and declining health, AI operators need to prove that the benefits and opportunities outweigh the extensive costs. AI data centres have real-world implications for nearby communities, but as projects are pushed through rapidly without robust impact assessments, whether those impacts are acceptable or sustainable remain unknown. Impact assessments also investigate the social, economic, and health impacts, providing a cost-benefit analysis to better weigh a given project. In their absence, the public is forced to take companies at their word, which is currently limited. If AI data centres are as beneficial, sustainable, and lucrative as their proponents would claim, then they have nothing to fear from a robust impact assessment. The process would serve to increase transparency, proactively identify issues and address public concerns, and foster local goodwill. AESO maintains an interactive Connections Project List Dashboard that displays all projects in development looking to connect to the provincial grid, including data centres. Projects can be sorted by type and size by clicking on the table headings. June 10, 2026 Alberta and Canada have stated that multiple Hyperscale Artificial Intelligence (AI) data centres will not...